HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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Abstract. A Hamiltonian bundle $M \hookrightarrow P \to X$ (with compact monotone fibers) induces, via Floer theory, a type of "bundle of A_∞ categories" over X, with fiber the Fukaya category of M. Morita theory of A_∞ categories, the above picture for $X = S^m$, and geometric representation theory yield the following: if G is a compact connected Lie group and R is a commutative ring, then the construction gives natural homomorphisms

$$\pi_m(BG) \to K_m^{Cat}(R),$$

where $K_m^{Cat}(R)$ are categorified algebraic K-theory groups of R, analogous to Toën's secondary K-theory. Furthermore, the homomorphisms are induced by a natural homotopy class $[B\,\mathrm{Ham}(G/T) \to K^{Cat}(R)]$, where $K^{Cat}(R)$ is the associated spectrum. We also construct underlying maps of this type to classical algebraic K-theory of R. This framework gives a geometry-powered proof that $K_2^{Cat}(\mathbb{Z})$ is infinitely generated (with the details to appear in a future work). This is in contrast to Quillen's finite generation result for standard algebraic K-theory of $\mathbb{Z}$. Taking the Langlands dual of G, we explore a conjectural relationship between the images of the corresponding homomorphisms above.

1. Introduction

A guiding idea is that a Hamiltonian bundle $M \hookrightarrow P \to Y$ gives rise to a kind of bundle of A_∞ categories, with fiber the Fukaya category of M. Precisely what this means is an interesting mathematical problem in its own right, and one answer is developed in [21]. Notably, these are not local systems; as shown in [19], the resulting bundles are generally non-trivial over spheres.

The second idea, due to Toën [26], is that there is a refinement of algebraic K-theory of a commutative ring R obtained by replacing R-modules with dg or A_∞ categories over R. We show that these two ideas naturally fit together. We use the above bundle assignment, with $Y = S^m$, to associate to each such P an element in categorified and standard algebraic K-theory of R. In particular, we get a geometric insight into algebraic K-theory of $\mathbb{Z}$, especially its categorified variant $K_m^{Cat}(\mathbb{Z})$. For example, we obtain an infinite generation result for $K_2^{Cat}(\mathbb{Z})$. At the end, we explore connections with Langlands duality. We thus link algebraic K-theory, Floer theory, and aspects of geometric representation theory in one framework.

Although in the main examples here the principal actors are compact groups, and in particular $PU(n)$, the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds. We quickly introduce the latter. Let (M, ω) be a symplectic manifold, assumed throughout to be compact, where ω is a closed nondegenerate 2-form on M. The group of diffeomorphisms ϕ of M, satisfying $\phi^*\omega = \omega$, is called the symplectomorphism group of (M, ω). When M is simply connected, the identity component of it is also the group of Hamiltonian symplectomorphisms, which we denote by $\mathcal{H} = \mathrm{Ham}(M, \omega)$. In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields $\{X_t\}$, $t \in [0, 1]$, satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions $\{H_t\}$ on M.

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The group $\mathcal{H}$ is one of the more enigmatic objects of mathematics. It is always (unless $M = pt$) an infinite-dimensional Fréchet Lie group with its C^∞ topology. But it has a natural, bi-invariant Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics; see, for instance, Polterovich [15].

Much of the algebraic/topological structure of $\mathcal{H}$ is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration* $M \hookrightarrow P \to Y$ is a smooth fiber bundle over Y, with fiber a symplectic manifold (M, ω), whose structure group is contained in $\mathrm{Ham}(M, \omega)$. For example, take $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$, where ω_{FS} is the Fubini-Study symplectic form. From now on, we omit ω_{FS} from all notation; it is implicitly always the chosen symplectic form on $\mathbb{CP}^n$. Suppose, for instance, that the structure group of $\mathbb{CP}^n \hookrightarrow P \to Y$ is $G = PU(n+1) \subset \mathrm{Ham}(\mathbb{CP}^n)$.

The other ingredient is algebraic K-theory, which associates an infinite loop space[1] $K(R)$ to any ring R, and hence abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called K-theory groups of R. This is a powerful tool in algebraic topology and ring theory, with applications for instance in number theory and geometric topology.

One motivating example for algebraic K-theory is the following. If R is the non-commutative ring $\mathcal{K}$ of compact operators on a separable complex Hilbert space, then the algebraic K-theory space $K(\mathcal{K})$[2] is identified with the topological K-theory space $KU = BU \times \mathbb{Z}$; see Karoubi [9], Suslin-Wodzicki [25]. Thus a complex vector bundle E over a space X determines a homotopy class $[f_E : X \to K(\mathcal{K})]$, which we might call a *geometric generator* of the corresponding abelian group. We shall show that in a certain sense this extends to a general ring R, provided R is commutative and we replace complex vector bundles by Hamiltonian fibrations.

For each commutative ring R, we construct an infinite loop space $K^{Cat}(R)$ called categorified algebraic K-theory of R. The groups $K^{Cat}_m(R) = \pi_m(K^{Cat}(R))$ can also be understood to be generated by the aforementioned bundles of A_∞ categories over S^m, with fibers A_∞ categories of finite type over R; see Definition 3.1. In principle, this categorified algebraic K-theory of R is only a slight variation of the secondary K-theory of R introduced by Toën, with the main difference being that $\mathbb{Z}$-graded chain complexes are replaced by $\mathbb{Z}_2$-graded chain complexes.
Standing assumption. The finite type condition in the concrete examples of the paper reduces to the condition that the Fukaya A_∞ algebra of the central monotone torus fiber in a closed toric monotone symplectic manifold is finite type over R. This will be our conjectural assumption. Thus some of the results below, most notably Claim 1.2, are conditional on this. This assumption holds for example if the homological mirror symmetry holds over R in the monotone toric case, as finiteness then follows from finiteness of the corresponding Landau–Ginzburg model. In the case of $M = \mathbb{CP}^n$, the assumption is known to hold for concrete examples of R, see Proposition 4.7.

The following is Theorem 4.5 combined with the standing assumption above. Since the above discussion mentioned a smooth bundle, we note that the universal bundles over $B\,\mathrm{Ham}(M, \omega)$, BG, etc. are smooth in needed sense for [21]; this is based on the language of smooth simplicial sets [22].

Theorem 1.1. *If* $\mathrm{Fuk}(M; R)$ *is finite type, then the data of the isomorphism class of* $M \hookrightarrow P \to Y$ *determines a class*

$$[Y \to K^{Cat}(R)].$$

[1]In this paper, the term infinite loop space is meant to be synonymous with connective Ω-spectrum, i.e. a sequence of spaces $\{E_n\}_{n \in \mathbb{N}}$ with $E_n \simeq \Omega E_{n+1}$. Then $K(R)$ is the notation for the 0-space E_0 of the corresponding spectrum.

[2]Rather one must take a certain non-connective variant of the algebraic K theory construction.

In particular, taking the associated $\mathbb{CP}^n$ *bundle over* $BPU(n+1)$ *and* $B\,\mathrm{Ham}(\mathbb{CP}^n, \omega)$*, we get natural classes*

$$[BPU(n+1) \to K^{Cat}(\mathbb{Z})],\ [B\,\mathrm{Ham}(\mathbb{CP}^n, \omega) \to K^{Cat}(\mathbb{Z})],$$

with the first factoring through the second.

Let ϕ_m denote the maps on homotopy groups induced by the classes in the theorem above, so $\phi_m : \pi_m(B\,\mathrm{Ham}(\mathbb{CP}^n, \omega)) \to K_m^{Cat}(\mathbb{Z})$, and similarly for $BPU(n+1)$.

Example 1. In the case of $n = 1$, in [19], using geometric arguments we outline the proof of non-triviality of ϕ_4. This contrasts with a theorem of Rognes [17] on the triviality of $K_4(\mathbb{Z})$.

Claim 1.2. *For each prime p, the homomorphism*

$$\phi_2 : \mathbb{Z}_p \simeq \pi_2(BPU(p)) \to K_2^{Cat}(\mathbb{Z})$$

is injective. Consequently, $K_2^{Cat}(\mathbb{Z})$ *is infinitely generated.*

Since the groups $K_m(\mathbb{Z})$ are always finitely generated by the foundational results of Quillen [16], the claim is perhaps surprising. One geometric ingredient for the proof is the now classical part of symplectic geometry: the Seidel homomorphism [23]. The details will appear in a future work [20].

1.1. **Generalizing to G/T.** Theorem 1.1 extends to $M = G/T$, where G is compact and connected. However, the details in that case also involve non-elementary geometric representation theory, see Section 4.1. We get a natural class

$$[B\,\mathrm{Ham}(G/T) \to K^{Cat}(R)], \tag{1.3}$$

but after fixing a certain choice (a distinguished toric degeneration of G/T).

Remark 1.4. Although the Floer theory works perfectly well in our context over any R due to monotonicity, the finite type assumptions are in principle sensitive to R. For the construction of (1.3) the finite type requirement is reduced to having finite type of the Fukaya A_∞ algebra over R of the central compact monotone torus fiber of a completely integrable system on G/T corresponding to a distinguished toric degeneration. By our standing assumption this is finite type.

1.2. **Pushing forward to classical K-theory.** We can also pass to classical algebraic K-theory, instead of its categorified analogue. We take a 2-periodic variant of classical algebraic K-theory of R, which we denote by $K_m^{\mathbb{Z}_2}(R)$. Let $\mathrm{Perf}_R^{\mathbb{Z}_2}$ denote the Waldhausen category of perfect 2-periodic R-complexes, i.e. those 2-periodic complexes which are quasi-isomorphic to a 2-periodic complex

$$P^0 \to P^1 \to P^0$$

with P^0 and P^1 finitely generated projective R-modules. The weak equivalences are quasi-isomorphisms. Then $K^{\mathbb{Z}_2}(R)$ is the algebraic K-theory infinite loop space $K(\mathrm{Perf}_R^{\mathbb{Z}_2})$.

Remark 1.5. It was pointed out to me by Bertrand Toën that for R a regular ring, for example $R = \mathbb{Z}$, $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$. This can be proved with $\mathbb{A}^1$ homotopy theory, although we do not technically need this here.

The same basic framework of the paper yields the following.

Theorem 1.6. *For any compact connected Lie group G, after fixing a distinguished toric degeneration, there is a natural homotopy class*

$$[BG \to K^{\mathbb{Z}_2}(R)], \tag{1.7}$$

which factors through a natural class $[B\,\mathrm{Ham}(G/T) \to K^{\mathbb{Z}_2}(R)]$.

Taking the Langlands dual of G, we explore a correspondence between the respective classes in Section 6.

Question 1. Is it possible to obtain the classes

$$[BG \to K^{\mathbb{Z}_2}(R)],\ [BG \to K^{Cat}(R)]$$

by techniques of homotopy theory and algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

It should be noted that compared to the maps ϕ_m, showing nontriviality of (1.7) is harder (and is an open problem), even restricting to $PU(n)$ and $R = \mathbb{Z}$. This might seem paradoxical, but the idea is that $K^{Cat}_m(R)$ sees more information, so it is easier to find nontrivial things in the image of ϕ_m.

2. Preliminaries

We will mostly follow Keller [10], but the base commutative ring will be denoted R to avoid overloading notation. An A_∞ category C over R is a form of non-associative dg-category, with $hom_C(a,b)$ a cochain complex over R, and strict associativity replaced by homotopy coherent associativity. We refer the reader to standard references for further details; see, for instance, [14], which will be used later.

Denote by $A_\infty \operatorname{Cat}^{strict}_R$ the category of strictly unital, $\mathbb{Z}$-graded A_∞ categories whose morphisms are strict unital A_∞ functors. We set $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$ to be analogous to $A_\infty \operatorname{Cat}^{strict}_R$, but with chain complexes $\mathbb{Z}_n$-graded instead of $\mathbb{Z}$-graded. This means that all structure maps, functors, etc., are with respect to the $\mathbb{Z}_n$-grading. Various subscripts R may be omitted.

There is an endofunctor

$$T : \mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}} \to \mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}},$$

such that $T(B)$ is idempotent complete and pretriangulated for each B. We may denote by $\widehat{C}$ the category $T(C)$.

Let $\operatorname{Cat}^{\mathbb{Z}_n}_R$ denote the category of small R-linear $\mathbb{Z}_n$-graded categories. For $C \in \mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$, set $hC \in Cat^{\mathbb{Z}_n}_R$ to be the category with the same objects and with $hom_{hC}(a,b) = H^*(hom_C(a,b))$.

Given a morphism $f : A \to B$ in $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$, we denote by $f_* : hA \to hB$ the induced morphism in $Cat^{\mathbb{Z}_n}_R$. We say that f is a *quasi-equivalence* if f_* is an equivalence. We will say that f is a *Morita equivalence* if the induced functor $T(A) \to T(B)$ is a quasi-equivalence. We denote by $Hmo^{\mathbb{Z}_2}$ the homotopy category of $A_\infty \operatorname{Cat}^{strict,\mathbb{Z}_2}_R$ with respect to Morita equivalences. As explained by Keller, this is a pointed category with the zero object: the category 0 with one object and zero endomorphism complex.

The category $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$ is cocomplete, see [14]. Furthermore, it has a model category structure [13]. Applying left Bousfield localization along Morita equivalences we get what is often called the Morita model structure. However, we do not explicitly need the language of model structures. The main algebraic ingredient is the following.

Definition 2.1. A ***short exact sequence*** in $A_\infty \operatorname{Cat}^{strict,\mathbb{Z}_2}_R$ is a sequence $A \xrightarrow{I} B \xrightarrow{P} C$, such that in $Hmo^{\mathbb{Z}_2}$, I is the kernel of P and P is the cokernel of I.

As explained by Keller in [10], an explicit example for such a short exact sequence is given by a Drinfeld dg-quotient sequence $A \hookrightarrow B \to B/A$.

2.1. **Simplicial sets.** We will need basic theory of simplicial sets, particularly the notion of a simplex category, for the definition of bundles of A_∞ categories. We denote by Δ the simplex category:

- The set of objects of Δ is $\mathbb{N}$.
- $\hom_\Delta(n, m)$ is the set of non-decreasing maps $[n] \to [m]$, where $[n] = \{0, 1, \dots, n\}$, with its natural order.

A simplicial set X is a functor
$$X : \Delta^{op} \to Set.$$
$X(n)$ is the set of n-simplices of X. Δ^d will denote a particular simplicial set: the standard representable d-simplex, with
$$\Delta^d(n) = hom_\Delta(n, d).$$
Morphisms of simplicial sets are given by natural transformations of functors, and these objects and morphisms form a category denoted by $sSet$.

An important ingredient in our construction will be the following.

Definition 2.2. For X a simplicial set, $\Delta(X)$ will denote the ***simplex category of*** X. This is the category such that:

- The set of objects $\operatorname{obj}\Delta(X)$ is the set of simplicial maps:
$$\Sigma : \Delta^d \to X, \quad d \geq 0.$$
- Morphisms $f : \Sigma_1 \to \Sigma_2$ are commutative diagrams in $sSet$:

$$\begin{tikzcd} \Delta^d \arrow[r] \arrow[dr, "\Sigma_1"] & \Delta^n \arrow[d, "\Sigma_2"] \\ & X. \end{tikzcd} \tag{2.3}$$

Given a morphism $f : X \to Y$ of simplicial sets, we set $\Delta f : \Delta(X) \to \Delta(Y)$ to be the functor defined on objects by $\Delta f(\Sigma) = f \circ \Sigma$. If $m : \Sigma_1 \to \Sigma_2$ is a morphism in $\Delta(X)$:

$$\begin{tikzcd} \Delta^k \arrow[r, "m"] \arrow[dr, "\Sigma_1"] & \Delta^d \arrow[d, "\Sigma_2"] \\ & X, \end{tikzcd}$$

then obviously the diagram below also commutes:

$$\begin{tikzcd} \Delta^k \arrow[r, "m"] \arrow[dr, "h_1"] & \Delta^d \arrow[d, "h_2"] \\ & Y, \end{tikzcd}$$

where $h_i = \Delta f(\Sigma_i) = f \circ \Sigma_i$, $i = 1, 2$. Thus, the latter diagram determines a morphism $\Delta f(m) : h_1 \to h_2$ in $\Delta(Y)$. Clearly, this determines a functor Δf as needed.

Definition 2.4. Let X be a simplicial set, and C any category. Let $F_i : \Delta(X) \to C$, $i = 0, 1$ be functors. A ***concordance*** of F_0 to F_1 is a functor
$$\widetilde{F} : \Delta(X \times \Delta^1) \to C$$
such that $\widetilde{F}|_{\Delta(X \times \{0\})} = F_0$ and $\widetilde{F}|_{\Delta(X \times \{1\})} = F_1$, where $\{0\}, \{1\}$ here denote the images of the natural endpoint inclusions $\Delta^0 \to \Delta^1$.

3. Definition of the categorified algebraic K-theory groups of R

We need to impose some algebraic finiteness conditions on our A_∞ categories. Otherwise, any K-theory we construct would be trivial due to the Eilenberg swindle; the same reason that K-theory of infinite-dimensional vector bundles is trivial. A natural requirement is that our A_∞ categories be smooth and proper, or saturated in the sense of Kontsevich-Soibelman [12], since this is a dualizability condition; see [27]. In the $\mathbb{Z}_n$-graded setting, the analogue of this condition is more complex, so we will instead work with a less restrictive notion, which is not too hard to check for geometric examples.

Set $\mathcal{A} = \mathcal{A}_R \subset A_\infty Cat_R^{\mathbb{Z}_2, strict}$ to be the full subcategory whose objects are of finite type, where the latter condition is defined as follows.

Definition 3.1. Let B be a $\mathbb{Z}_2$-graded A_∞-category over R. We say that B is of finite type if the following two conditions hold.

(1) B is Morita equivalent to an A_∞-algebra A, whose underlying chain complex is in $\mathrm{Perf}_\mathrm{R}^{\mathbb{Z}_2}$. Here $\mathrm{Perf}_\mathrm{R}^{\mathbb{Z}_2}$ consists of those 2-periodic complexes which are quasi-isomorphic to a 2-periodic complex
$$P^0 \to P^1 \to P^0$$
with P^0 and P^1 finitely generated projective R-modules.

(2) The Hochschild chain complex $Hoch_\bullet(B)$ is in $\mathrm{Perf}_\mathrm{R}^{\mathbb{Z}_2}$.

If R is not specified, and we say finite type then $R = \mathbb{Z}$ is assumed, as by base change this implies finite type over any other R.

Set $\widehat{\mathcal{A}}_R \subset \mathcal{A}_R$ to be the full subcategory with objects pretriangulated and idempotent complete. As outlined by Toën in [26], $\widehat{\mathcal{A}}_R \subset \mathcal{A}_R$ has the structure of a Waldhausen category. The cofibrations are taken to be fully-faithful, strict, unital, injective on sets of objects A_∞ functors. Weak equivalences are Morita equivalences. Strictly speaking, this is not a Waldhausen category, but rather ∞-Waldhausen in the sense of Barwick [2], since, for example, the zero category 0 (the category with one object and zero complex as endomorphism set) is not initial in $\widehat{\mathcal{A}}_R$ but the hom-sets $hom_{\widehat{\mathcal{A}}_R}(0, A)$ are contractible when taken in the ∞-localization of $\widehat{\mathcal{A}}_R$ with respect to the Morita equivalences.

Given this, we define
$$K_m^{Cat}(R) = \pi_m K(\widehat{\mathcal{A}}_R),$$
where $K(\widehat{\mathcal{A}}_R)$ is the associated Waldhausen spectrum. Calculations from this general perspective are difficult, so we now describe a more concrete presentation.

3.1. Concrete presentation of $K_m^{Cat}(R)$.

Definition 3.2. Let X be a simplicial set. A ***bundle of*** A_∞ ***categories over*** X is a functor $F : \Delta(X) \to w\widehat{\mathcal{A}}_R$. F will be called a ***bundle functor over*** X.

Suppose we have bundle functors over X, fitting into a sequence of natural transformations of functors to $\widehat{\mathcal{A}}_R$ (as opposed to $w\widehat{\mathcal{A}}_R$):
$$F_0 \to F_1 \to F_2, \tag{3.3}$$
such that for each object $\Sigma \in \Delta(X)$ the resulting sequence
$$F_0(\Sigma) \to F_1(\Sigma) \to F_2(\Sigma), \tag{3.4}$$
is a short exact sequence in $\widehat{\mathcal{A}}_R$. Then we call $F_0 \to F_1 \to F_2$ a ***short exact sequence***.

Definition 3.5. Let X be a Kan complex. Set $B(X,R)$ to be the free abelian group generated by the set of concordance classes $[F]$ of functors $F : \Delta(X) \to w\widehat{\mathcal{A}}_R$. We define $K^{Cat}(X,R)$ to be the quotient of $B(X,R)$ by the subgroup generated by

$$\{[F_0] - [F_1] + [F_2] \,|\, \text{there is a short exact sequence } F_0 \to F_1 \to F_2\}.$$

This is clearly contravariantly functorial in X. Thus, if $f : X \to Y$ is a simplicial map, then we get a group homomorphism $f^* : K^{Cat}(Y,R) \to K^{Cat}(X,R)$ defined on generators by $f^*([F]) = [F \circ \Delta f]$.

The groups $K^{Cat}(X,R)$ are covariantly functorial in R by "extension of scalars". Furthermore, $X \mapsto K^{Cat}(X,R)$ gives a contravariant homotopy functor, because by the definition of concordance, if $f : X \to Y$ is homotopic to g, then $f^* = g^*$. In particular, if X_0 is homotopy equivalent to X_1 then $K^{Cat}(X_0,R) \simeq K^{Cat}(X_1,R)$.

Let $S^n_\bullet$ be the singular complex of S^n.

Definition 3.6. For $n > 0$, we define

$$K^{Cat,g}_n(R) = \ker\{(\Delta^0 \to S^n_\bullet)^* : K^{Cat}(S^n_\bullet, R) \to K^{Cat}(\Delta^0, R)\},$$

and define

$$K^{Cat,g}_0(R) = K^{Cat}(\Delta^0, R).$$

In other words, $K^{Cat,g}_0(R)$ is the quotient of the free abelian group generated by Morita equivalence classes $[C]$, by the subgroup generated by:

$$\{[C_0] - [C_1] + [C_2] \,|\, \text{there is a short exact sequence } C_0 \to C_1 \to C_2\}.$$

The superscript g is meant to evoke that these groups are constructed via a concrete geometric presentation.

Let

$$\mathfrak{i} : |w\widehat{\mathcal{A}}_R| \to K^{Cat}(R)$$

denote the canonical map from the space of weak equivalences to the Waldhausen K-theory space.

To see the relationship between $K^{Cat,g}_n(R)$ and $K^{Cat}_n(R)$, note that for a Kan complex X, a bundle functor $F : \Delta(X) \to w\widehat{\mathcal{A}}_R$ induces a sequence

$$|X| \simeq |\Delta(X)| \xrightarrow{|F|} |w\widehat{\mathcal{A}}_R| \xrightarrow{\mathfrak{i}} K^{Cat}(R), \tag{3.7}$$

where $|X|$ also denotes the geometric realization of X. Clearly, a concordance between F_0, F_1 induces a homotopy between $\mathfrak{i} \circ |F_i|$.

Let $hMaps(|X|, K^{Cat}(R))$ be the set of free homotopy classes $[|X| \to K^{Cat}(R)]$. Since $K^{Cat}(R)$ is an infinite loop space, $hMaps(|X|, K^{Cat}(R))$ is naturally an abelian group. Thus we get an induced group homomorphism:

$$\widetilde{\mathcal{J}}_X : B(X,R) \to hMaps(|X|, K^{Cat}(R)), \tag{3.8}$$

induced by the map $[F] \mapsto [\mathfrak{i} \circ |F|]$.

Claim 3.9. *The map $\widetilde{\mathcal{J}}_X$ passes to the quotient, inducing a map*

$$\mathcal{J}_X : K^{Cat}(X,R) \to hMaps(|X|, K^{Cat}(R)). \tag{3.10}$$

Furthermore, $\mathcal{J}_X$ is a group isomorphism, and induces a natural isomorphism $\mathcal{J}$ from the Ab-valued functor $X \mapsto K^{Cat}(X,R)$ to the functor $X \mapsto hMaps(|X|, K^{Cat}(R))$.

The proof of the claim will appear elsewhere, as it involves various algebraic topology details. Note that in general one cannot always concretely present Waldhausen K-theory groups in this manner. The proof in our setting hinges on the fact that the ∞-localization of $\widehat{\mathcal{A}}_R$ with respect to Morita equivalences has a concrete model $\mathcal{M}$, such that concordance classes of functors $F : \Delta(X) \to w\widehat{\mathcal{A}}_R$ are in correspondence with homotopy classes $[X \to Core(\mathcal{M})]$, where $Core(\mathcal{M})$ is the maximal Kan subcomplex. As remarked in the paragraph following Definition 3.1, this $\mathcal{M}$ is the ∞-Waldhausen category from which the spectrum $K^{Cat}(R)$ is constructed.

Given the claim, there is clearly an induced map

$$\mathcal{J}_m : K^{Cat,g}_m(R) \to \pi_m(K^{Cat}(R)) = K^{Cat}_m(R) \tag{3.11}$$

and it is a group isomorphism.

3.2. **A very basic computation.** We recall the definition of $K_0(R)$. This is the quotient of the free abelian group generated by the set of isomorphism classes of projective finitely generated R-modules, by the subgroup generated by

$$\{[A] - [B] + [C] \,|\, \text{exists an exact sequence } 0 \to A \to B \to C \to 0.\}$$

If $\mathcal{E}$ is a triangulated category, one defines $K_0(\mathcal{E})$ to be the quotient of the free abelian group, generated by the set of isomorphism classes in $\mathcal{E}$, by the subgroup generated by

$$\{[A] - [B] + [C] \,|\, \text{exists a distinguished triangle } A \to B \to C \to A[1]\}.$$

It is known that $K_k(D^{perf}(Mod_R)) \simeq K_k(R)$, where Mod_R denotes the category of projective finitely generated R-modules and $D^{perf}(Mod_R)$ denotes the derived category of perfect complexes.

Let $\mathcal{B}$ be an essentially small abelian category with enough projectives and finite global dimension, and let $D_n(\mathcal{B})$ denote its n-periodic derived category. We shall use the following theorem of Saito [18].

Theorem 3.12 (Saito)**.** *For n even, the natural homomorphism*

$$K_0(\mathcal{B}) \to K_0(D_n(\mathcal{B}))$$

is an isomorphism. Its inverse is the Euler-characteristic homomorphism

$$\chi_n : K_0(D_n(\mathcal{B})) \to K_0(\mathcal{B}) \tag{3.13}$$

$$\chi_n([V]) = \sum_{i=0}^{n-1} (-1)^i [H^i(V)]. \tag{3.14}$$

We apply this below with $\mathcal{B} = \mathrm{mod}_{\mathbb{Z}}$, the abelian category of finitely generated abelian groups. Thus $K_0(\mathcal{B}) = G_0(\mathbb{Z})$. Since $\mathbb{Z}$ is regular, the Cartan map identifies $K_0(\mathbb{Z})$ with $G_0(\mathbb{Z})$.

There is a group homomorphism

$$\phi : K^{Cat}_0(\mathbb{Z}) \to K_0(D_2(\mathrm{mod}_{\mathbb{Z}}))$$

induced by $[C] \mapsto [Hoch_\bullet(C)]$, where $Hoch_\bullet(C)$ is the Hochschild chain complex of C. For finite type C over $\mathbb{Z}$, $Hoch_\bullet(C) \in \mathrm{Perf}^{\mathbb{Z}_2}_{\mathbb{Z}}$, and hence it determines an object of $D_2(\mathrm{mod}_{\mathbb{Z}})$. The map ϕ is well-defined because the Hochschild functor $Hoch_\bullet$ is "localizing," which means in this case that a short exact sequence

$$C_0 \to C_1 \to C_2$$

in $\mathcal{A}_{\mathbb{Z}}$ is taken to an exact triangle

$$Hoch_\bullet(C_0) \to Hoch_\bullet(C_1) \to Hoch_\bullet(C_2) \to Hoch_\bullet(C_0)[1],$$

in the derived category $D_2(\mathrm{mod}_{\mathbb{Z}})$; see Keller [10, Theorem 5.2].

Corollary 3.15. *The composition map*

$$K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(\mathrm{mod}_{\mathbb{Z}})) \xrightarrow{\chi_2} G_0(\mathbb{Z}) \simeq K_0(\mathbb{Z}),$$

is non-trivial.

Proof. Take C to be a finite type $\mathbb{Z}_2$-graded A_∞ algebra over $\mathbb{Z}$ whose Hochschild homology $HH_\bullet(C)$, the homology of $Hoch_\bullet(C)$, has $HH_0(C)$ of rank 2 and $HH_1(C)$ torsion. In $G_0(\mathbb{Z})$, every finite torsion abelian group has class zero: for $\mathbb{Z}/n$, the exact sequence

$$0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n \to 0$$

gives $[\mathbb{Z}/n] = [\mathbb{Z}] - [\mathbb{Z}] = 0$, and every finite abelian group is a finite direct sum of such cyclic groups. Thus $[A] = (\mathrm{rk}\, A)\, [\mathbb{Z}]$ in $G_0(\mathbb{Z})$ for every finitely generated abelian group A, and hence, by (3.13),

$$\chi_2([Hoch_\bullet(C)]) = [HH_0(C)] - [HH_1(C)] = 2\, [\mathbb{Z}] \neq 0 \in G_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

As a geometric example, take $C = CF^\bullet(L_0, L_0)$, the Floer cochain algebra of the equator $L_0 \subset S^2$. For this toric Clifford-type model, the Hochschild homology is known via homological mirror symmetry calculations [24]. At least this is the case over field coefficients. In particular, tensoring with $\mathbb{Q}$, we get:

$$HH_\bullet(C) \otimes \mathbb{Q} \;\cong\; QH^\bullet(S^2; \mathbb{Q}) \;\cong\; H^\bullet(S^2; \mathbb{Q}),$$

whose even and odd ranks are 2 and 0, respectively. □

With the definition of $\mathrm{Fuk}^{L_0}(S^2, \omega)$ in the paragraph preceding Proposition 4.7, a further corollary of the proof above is:

Corollary 3.16. $[\mathrm{Fuk}^{L_0}(S^2, \omega)]$ *represents a non-trivial element in* $K_0^{Cat}(\mathbb{Z})$.

One natural question is how to produce non-trivial elements in $K_m^{Cat}(R)$ for higher m. We will construct such elements using Hamiltonian fibrations and Floer theory.

4. Bundles of A_∞ categories from Hamiltonian fibrations

The main construction of [21] takes as input a smooth Hamiltonian fibration $M \hookrightarrow P \to Y$, with (M, ω) a compact, monotone symplectic manifold, e.g. $\mathbb{CP}^n$. More generally, a monotone symplectic manifold (M, ω) is one satisfying $[\omega] = \lambda \cdot c_1(TM)$, for some $\lambda > 0$, viewed as functionals $\pi_2(M) \to \mathbb{R}$.

Denote by $Y_\bullet$ the simplicial set of smooth maps $\Delta^n \to Y$, where Δ^n also denotes the standard topological n-simplex.

The output is a functor

$$F_P : \Delta(Y_\bullet) \to wA_\infty Cat_R^{\mathbb{Z}_2, strict}, \tag{4.1}$$

whose concordance class is independent of auxiliary choices (these mainly consist of choices of almost complex structures and Hamiltonian connections.)[3]

For $x : \Delta^0 \to Y$ a point map, we have $F_P(x) = \mathrm{Fuk}(P_x, \omega_x)$, P_x denoting the fiber over x.

We can outline this further: if $\Sigma : \Delta^1 \to Y$ is a path from x to y, then $F_P(\Sigma)$ has the set of objects

$$\mathrm{obj}\, F_P(x) \sqcup \mathrm{obj}\, F_P(y).$$

[3] In [21] we work with $R = \mathbb{Q}$ but this was only used for inverting quasi-equivalences. As we don't need to do this here, the construction works over any commutative ring.

For $L_x \in \operatorname{obj} F_P(x)$ and $L_y \in \operatorname{obj} F_P(y)$, $hom_{F_P(\Sigma)}(L_x, L_y)$ is a certain chain complex generated by flat sections of an auxiliary Hamiltonian connection on $\Sigma^* P$, with endpoints of the sections on L_x and L_y.[4]

More generally, for a smooth $\Sigma : \Delta^n \to Y$ denote by $v_i : \Delta^0 \to Y$, $0 \leq i \leq n$, the corresponding vertex maps. Then $F_P(\Sigma)$ has the set of objects

$$\sqcup_i \operatorname{obj} F_P(v_i). \tag{4.2}$$

The hom sets are defined analogously so that an edge inclusion morphism $\Sigma^1 \to \Sigma$ in $\Delta(Y_\bullet)$ induces a fully-faithful embedding (and quasi-equivalence) $F_P(\Sigma^1) \to F_P(\Sigma)$. By an edge inclusion we mean a morphism of the following type:

$$\begin{tikzcd} \Delta^1 \arrow[r] \arrow[dr, "\Sigma^1"] & \Delta^n \arrow[d, "\Sigma"] \\ & Y. \end{tikzcd} \tag{4.3}$$

The only deep part of the construction is then the definition of the A_∞ structure on $F_P(\Sigma)$, which must also be done so that functoriality of F_P is satisfied. This involves some novel geometric topology of the universal curve, but we will not attempt to outline this further, instead referring the reader to the original construction in [21].

To get the algebraic finiteness condition on the Fukaya category we may impose the following condition.

Definition 4.4. We say that a compact monotone symplectic manifold is of ***finite monotone type over*** R if its monotone Fukaya category (see [24] and also [21]) is finite type over R.

Let us set $\widehat{F}_P = T \circ F_P$, which is then a bundle functor over $Y_\bullet$. Now, by standard homotopy theory, $|\Delta(Y_\bullet)| \simeq Y$, where, for a category C, $|C|$ is shorthand for the geometric realization of the nerve. We then have the following corollary of [21, Theorem 6.6].

Theorem 4.5. *Let $M \hookrightarrow P \to Y$ be a Hamiltonian fibration with fiber (M, ω) of finite monotone type over R. Then the assignment*

$$[P] \mapsto [\widetilde{\mathcal{J}}_{Y_\bullet}([\widehat{F}_P])] \in hMaps(Y, K^{Cat}(R)), \tag{4.6}$$

is well defined, where $[P]$ is the Hamiltonian isomorphism class of the fibration P, and $\widetilde{\mathcal{J}}$ is as in (3.8).

Working with the full monotone Fukaya category is impractical in our setting, especially for establishing finite type. We set $\mathrm{Fuk}^{L_0}(\mathbb{CP}^n)$ to be the full sub-category of the monotone Fukaya category of $\mathbb{CP}^n$, with objects Lagrangians Hamiltonian isotopic to L_0, where $L_0 \subset \mathbb{CP}^n$ is the Clifford torus, equipped with the trivial local system. Then we have:

Proposition 4.7. *Let R be a commutative ring in which $n+1$ is invertible. Then $\mathrm{Fuk}^{L_0}(\mathbb{CP}^n)$ is of finite type over R.*

Proof. It follows from the definition that $\mathrm{Fuk}^{L_0}(\mathbb{CP}^n)$ is quasi-equivalent to the A_∞-algebra

$$A = \hom_{\mathrm{Fuk}(\mathbb{CP}^n;R)}(L_0, L_0).$$

[4] These Hamiltonian connections are fixed per simplex, but need to satisfy compatibility conditions involving face maps of simplices.

The relevant toric Floer computation is local at the symmetric critical point of the Landau–Ginzburg potential

$$W = x_1 + \cdots + x_n + \frac{1}{x_1 \cdots x_n}.$$

At this point, the Hessian of W, written in logarithmic coordinates, is the matrix $I + J$, where J is the all-ones matrix. Thus

$$\det(I + J) = n + 1.$$

Since $n+1$ is invertible in R, this critical point is nondegenerate over R. By the standard computation of the Floer product for toric fibers, A is A_∞-quasi-isomorphic to the corresponding Clifford algebra determined by this nondegenerate Hessian.

It follows that the underlying R-complex of A is perfect. Moreover, the Hochschild chain complex $Hoch_\bullet(A)$ is a perfect 2-periodic R-complex. Therefore A, and hence the above orbit category, satisfies the finite-type condition over R. □

4.1. **G/T and toric degenerations.** Given a compact connected Lie group G, the maximal coadjoint orbit G/T can be made monotone with respect to the Kirillov-Kostant-Souriau symplectic structure ω_{KKS}, which for G/T is uniquely determined up to deformation equivalence, see for instance [11]. From now on, we omit this monotone ω_{KKS} from notation, but it will always be the implicit chosen symplectic form. We need a concrete finite type model for $\mathrm{Fuk}(G/T)$, where by model we don't necessarily mean Morita equivalent model.

If $G = PU(n)$ there is a distinguished monotone Lagrangian torus L_{GC} of $PU(n)/T$, arising as the central fiber of a Gelfand-Cetlin integrable system; see [6, Theorem B]. In a sense, this extends to a general compact connected G. In this case, G/T admits a flat toric degeneration to a toric variety $X(\Delta)$ associated with a string polytope Δ, see [4, 1]. This deformation defines a completely integrable system on G/T, with moment map to Δ [7]. By anchoring the construction to the standard string valuation, one identifies a distinguished toric degeneration where the resulting string polytope is Fano. Within this fixed framework, the fiber over the barycenter of Δ is a monotone Lagrangian torus that is well-defined up to Hamiltonian isotopy [5]. This torus will be denoted by $\mathcal{L}$. (Note that taking "non-standard" valuations, we may get exotic tori that are not Hamiltonian isotopic to $\mathcal{L}$ [5].)

We define $C_\mathcal{L} \subset \mathrm{Fuk}(G/T)$ to be the full subcategory whose objects are the orbit of $\mathcal{L}$ under the $\mathrm{Ham}(G/T)$ action. By our standing assumption as stated in the Introduction, this gives a finite type A_∞ category. We then construct a functor

$$F^\mathcal{L} : \Delta(B\,\mathrm{Ham}(G/T)_\bullet) \to w\mathcal{A}_R$$

as follows. Denote by $G/T \hookrightarrow P^U \to B\,\mathrm{Ham}(G/T)$ the associated universal bundle. For a point map $x : \Delta^0 \to B\,\mathrm{Ham}(G/T)$ we set $F^\mathcal{L}(x)$ to be the full subcategory of $\mathrm{Fuk}(P_x)$, with the set of objects $\phi_x(\mathrm{obj}\, C_\mathcal{L})$, where $\phi_x : G/T \to P_x$ is a frame over x (a $\mathrm{Ham}(G/T)$ structure group bundle map). Clearly the choice of frame is irrelevant, as changing the frame acts by an element of $\mathrm{Ham}(G/T)$, which preserves the orbit defining $C_\mathcal{L}$. The remaining construction of $F^\mathcal{L}$ is the same as the general construction of (4.1).

Given this $F^\mathcal{L}$ and its triangulated version $\widehat{F}^\mathcal{L} = T \circ F^\mathcal{L}$, following the proof of Corollary **??**, we again get a natural class:

$$[\phi^{\mathcal{L},G} : B\,\mathrm{Ham}(G/T) \to K^{Cat}(R)]. \tag{4.8}$$

5. Hamiltonian elements in classical algebraic K-theory of R

One of the goals is to obtain Hamiltonian elements in classical algebraic K-theory of R. Let $M \hookrightarrow P \to Y$ be a Hamiltonian fibration with compact monotone fiber (M, ω). Let

$$F_P : \Delta(Y_\bullet) \to wA_\infty Cat_R^{\mathbb{Z}_2, strict}$$

be as in the previous section. Then, under the finite-type assumption, the composition $Hoch_\bullet \circ F_P$ gives a functor to $w\,\mathrm{Perf}_R^{\mathbb{Z}_2}$. The Waldhausen K-theory infinite loop space $K(\mathrm{Perf}_R^{\mathbb{Z}_2})$ will be denoted by $K^{\mathbb{Z}_2}(R)$.

Restrict now to $Y = B\,\mathrm{Ham}(G/T)$, take $F^{\mathcal{L}}$ to be the functor defined as in the previous section, and $\Phi^{\mathcal{L}} = Hoch_\bullet \circ F^{\mathcal{L}}$. Then we get a map:

$$|\Phi^{\mathcal{L}}| : B\,\mathrm{Ham}(G/T) \to |w\,\mathrm{Perf}_R^{\mathbb{Z}_2}|,$$

whose homotopy class is well defined by the fact that the concordance class of $F^{\mathcal{L}}$ is well defined.

There is a natural map $|w\,\mathrm{Perf}_R^{\mathbb{Z}_2}| \to K^{\mathbb{Z}_2}(R)$, by generalities of the Waldhausen construction; see [28, Remark 8.3.2]. Composing with this map, we get a natural homotopy class:

$$[B\,\mathrm{Ham}(G/T) \to K^{\mathbb{Z}_2}(R)]. \tag{5.1}$$

Proof of Theorem 1.6. Compose the natural map

$$BG \to B\,\mathrm{Ham}(G/T)$$

with the class in (5.1). □

6. Connection to Langlands duality

It is natural to ask how Langlands duality fits into our story. Denote by $G^\vee$ the maximal compact subgroup of the Langlands dual $G_{\mathbb{C}}^\vee$ of the complexification $G_{\mathbb{C}}$. Let $G_{ad} = G/Z(G)$ be the adjoint form. If G is simply laced (e.g. $G = SU(n)$), the flag manifolds G/T and $G^\vee/T^\vee$ are identified. Under this identification, $G_{ad} \simeq G_{ad}^\vee$. Motivated by this:

Question 2. For each compact connected semisimple Lie group G, do the images of

$$(\phi^{\mathcal{L},G})_* : \pi_m(BG_{ad}) \to K_m^{Cat}(R)$$

and

$$(\phi^{\mathcal{L},G^\vee})_* : \pi_m(BG_{ad}^\vee) \to K_m^{Cat}(R)$$

coincide? Here $\phi^{\mathcal{L},G}$ denotes the restriction of the map (4.8), via the natural map $BG_{ad} \to B\,\mathrm{Ham}(G/T)$.

For $m = 0$, this is saying that $\mathrm{Fuk}(G/T)$ and $\mathrm{Fuk}(G^\vee/T)$[5] represent the same classes in $K_0^{Cat}(R)$. This assertion holds if these categories are Morita equivalent, but it is much weaker. The case $m = 0$ is then loosely comparable to the homological mirror symmetry conjecture of Kontsevich.

Note that the domains of $\phi^{\mathcal{L},G}$ and $\phi^{\mathcal{L},G^\vee}$ and the manifolds G/T and $G^\vee/T$ are generally very different. The idea is that the images in (categorified) algebraic K-theory depend only on some representation-theoretic data that is invariant under Langlands duality. Thematically, such a phenomenon already appears in electric-magnetic dualities, and this is one inspiration. Related work on topological aspects of Langlands duality/correspondence appears in [3]. The seminal reference for electric-magnetic duality vis-à-vis Langlands duality is Kapustin-Witten [8].

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[5]Actually their models $C_{\mathcal{L}}$ as in Section 4.1.

References

[1] V. Alexeev and M. Brion. Toric degenerations of spherical varieties. *Selecta Mathematica*, 10(4):453–470, 2004. doi:10.1007/s00029-004-0382-x. 4.1

[2] C. Barwick. On the algebraic K-theory of higher categories. *J. Topol.*, 9(1):245–347, 2016. doi:10.1112/jtopol/jtv042. 3

[3] D. Ben-Zvi and D. Nadler. The character theory of a complex group. *American Journal of Mathematics*, 135(6):1659–1714, 2013. doi:10.1353/ajm.2013.0046. 6

[4] P. Caldero. Toric degenerations of schubert varieties. *Transformation Groups*, 7:51–60, 2002. doi:10.1007/bf01253464. 4.1

[5] Y. Cho, M. Kim, Y. Kim, and E. Park. Cluster algebras and monotone Lagrangian tori. *Advances in Mathematics*, 480:110481, 2025. doi:10.1016/j.aim.2025.110481. 4.1

[6] Y. Cho and Y. Kim. Monotone lagrangians in flag varieties. *International Mathematics Research Notices*, 2021(18):13892–13945, 2021. doi:10.1093/imrn/rnz227. 4.1

[7] M. Harada and K. Kaveh. Integrable systems, toric degenerations and okounkov bodies. *Inventiones Mathematicae*, 202(3):927–985, 2015. 4.1

[8] A. Kapustin and E. Witten. Electric-magnetic duality and the geometric langlands program, 2007. URL: https://arxiv.org/abs/hep-th/0604151, arXiv:hep-th/0604151. 6

[9] M. Karoubi and M. Wodzicki. Algebraic k-theory of operator rings. *Journal of Geometry and Physics*, 146:103514, 2019. URL: https://www.sciencedirect.com/science/article/pii/S0393044019301962, doi:https://doi.org/10.1016/j.geomphys.2019.103514. 1

[10] B. Keller. On differential graded categories. *Proceedings of the international congress of mathematicians (ICM), Madrid, Spain, August 22–30, 2006. Volume II: Invited lectures. Zürich: European Mathematical Society (EMS)*, 2006. 2, 2, 3.2

[11] A. A. Kirillov. Lectures on the orbit method. *Providence, RI: American Mathematical Society*, 2004. 4.1

[12] M. Kontsevich and Y. Soibelman. Notes on A_∞-algebras, A_∞-categories and non-commutative geometry. In *Homological mirror symmetry. New developments and perspectives*, pages 153–219. Berlin: Springer, 2009. doi:10.1007/978-3-540-68030-7_6. 3

[13] M. Ornaghi. A model structure on the category of A_∞-categories with strict morphisms, 2025. URL: https://arxiv.org/abs/2506.22847, arXiv:2506.22847. 2

[14] M. Ornaghi. The Homotopy Theory of A∞categories. *International Mathematics Research Notices*, 2025(11):rnaf127, 05 2025. arXiv:https://academic.oup.com/imrn/article-pdf/2025/11/rnaf127/63393439/rnaf127.pdf, doi:10.1093/imrn/rnaf127. 2

[15] L. Polterovich. *The geometry of the group of symplectic diffeomorphism.* Lectures in Mathematics, ETH Zürich. Basel: Birkhäuser. xii, 132 p. sFr. 34.00; DM 44.00; öS 321.00 , 2001. 1

[16] D. Quillen. Higher algebraic k-theory i. In H. Bass, editor, *Algebraic K-Theory I: Higher K-Theories*, volume 341 of *Lecture Notes in Mathematics*, pages 85–147. Springer, 1973. doi:10.1007/BFb0067053. 1

[17] J. Rognes. $K_4(\mathbb{Z})$ is the trivial group. *Topology*, 39(2):267–281, 2000. doi:10.1016/S0040-9383(99)00007-5. 1

[18] S. Saito. A note on Grothendieck groups of periodic derived categories. *J. Algebra*, 609:552–566, 2022. doi:10.1016/j.jalgebra.2022.06.036. 3.2

[19] Y. Savelyev. Global Fukaya category II. *http://yashamon.github.io/web2/papers/fukayaII.pdf*. 1, 1

[20] Y. Savelyev. Infinite generation of the categorifed algebraic K-theory of $\mathbb{Z}$. *in construction*. 1

[21] Y. Savelyev. Global Fukaya category I. *Int. Math. Res. Not.*, 2023(21):18302–18386, 2023. doi:10.1093/imrn/rnad013. 1, 1, 4, 3, 4, 4.4, 4

[22] Y. Savelyev. Smooth simplicial sets and universal chern–weil for infinite-dimensional groups. *Journal of the London Mathematical Society*, 113(1):e70417, 2026. URL: https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/jlms.70417, arXiv:https://londmathsoc.onlinelibrary.wiley.com/doi/pdf/10.1112/jlms.70417, doi:https://doi.org/10.1112/jlms.70417. 1

[23] P. Seidel. π_1 of symplectic automorphism groups and invertibles in quantum homology rings. *Geom. Funct. Anal.*, 7(6):1046–1095, 1997. doi:10.1007/s000390050037. 1

[24] N. Sheridan. On the Fukaya category of a Fano hypersurface in projective space. *Publ. Math., Inst. Hautes Étud. Sci.*, 124:165–317, 2016. doi:10.1007/s10240-016-0082-8. 3.2, 4.4

[25] A. A. Suslin and M. Wodzicki. Excision in algebraic k-theory. *Annals of Mathematics*, 136(1):51–122, 1992. URL: http://www.jstor.org/stable/2946546. 1

[26] B. Toën. The homotopy theory of dg-categories and derived Morita theory. *Invent. Math.*, 167(3):615–667, 2007. 1, 3

[27] B. Toën. *Lectures on dg-categories.* Berlin: Springer, 2011. doi:10.1007/978-3-642-15708-0_5. 3

[28] C. A. Weibel. *The K-book. An introduction to algebraic K-theory*, volume 145 of *Grad. Stud. Math.* Providence, RI: American Mathematical Society (AMS), 2013. 5

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